

Figure 1 — Generation Success Rate Comparison

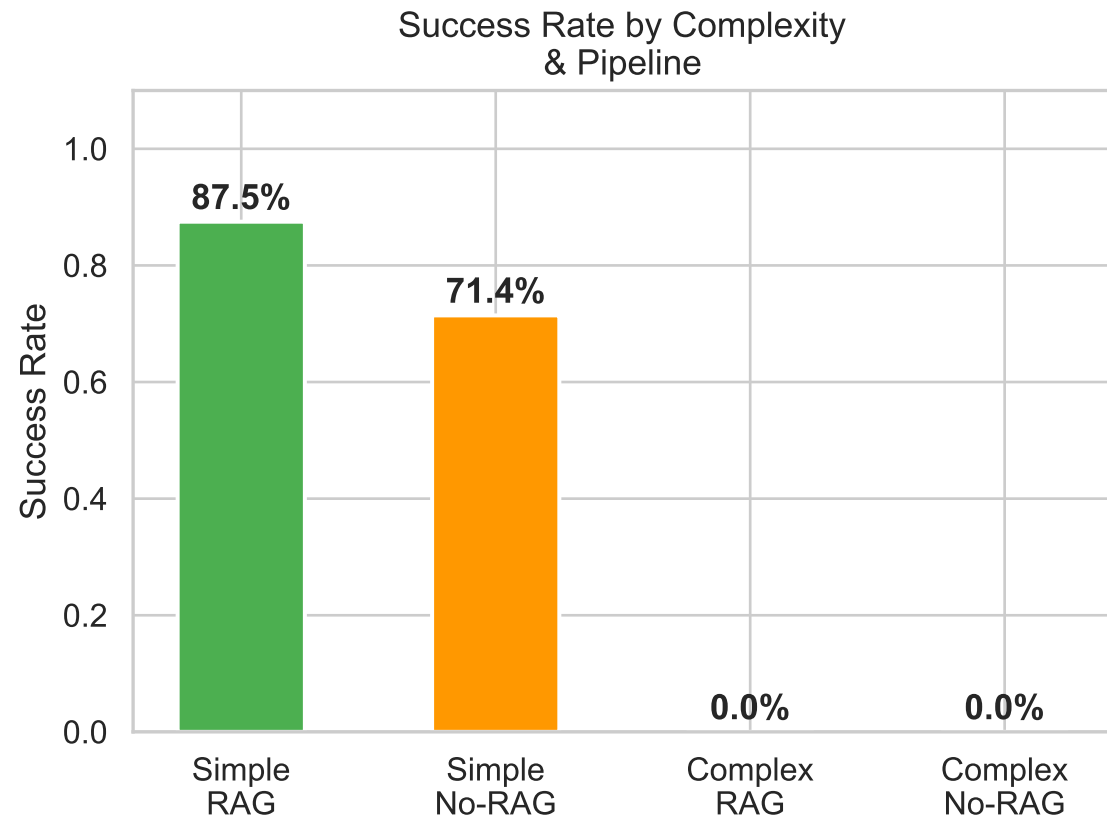
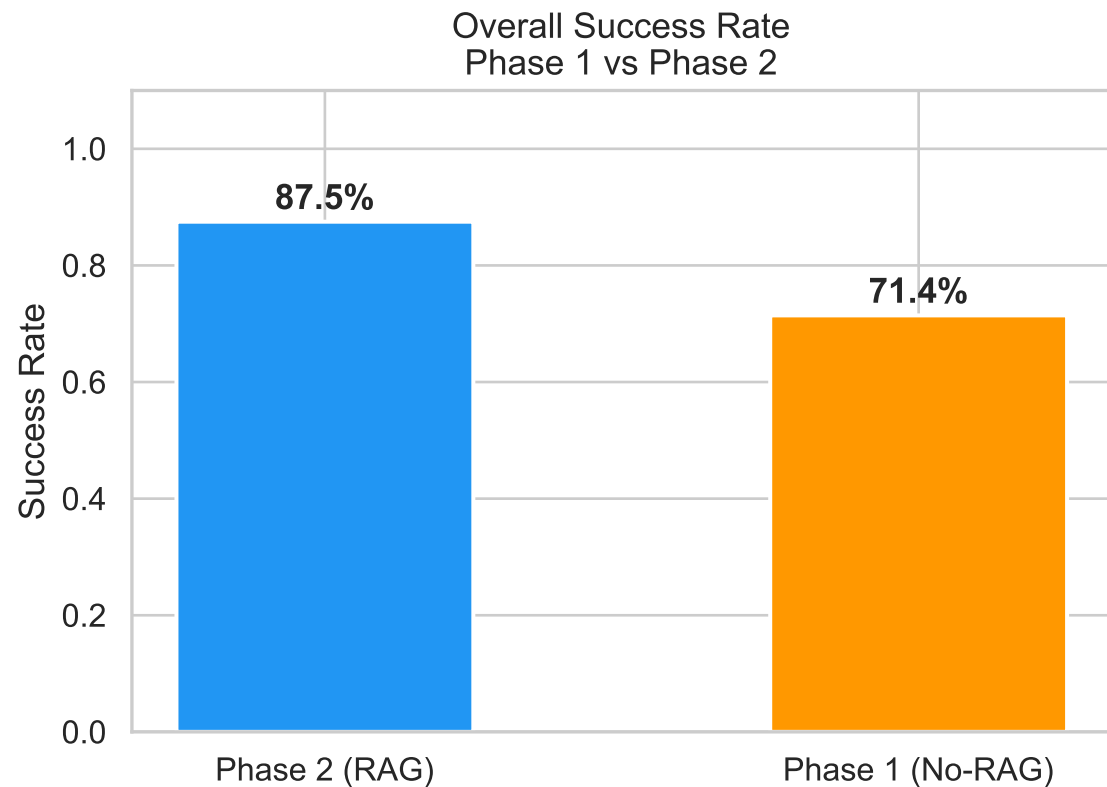


Figure 2 — VLM Score Distribution by Pipeline & Complexity

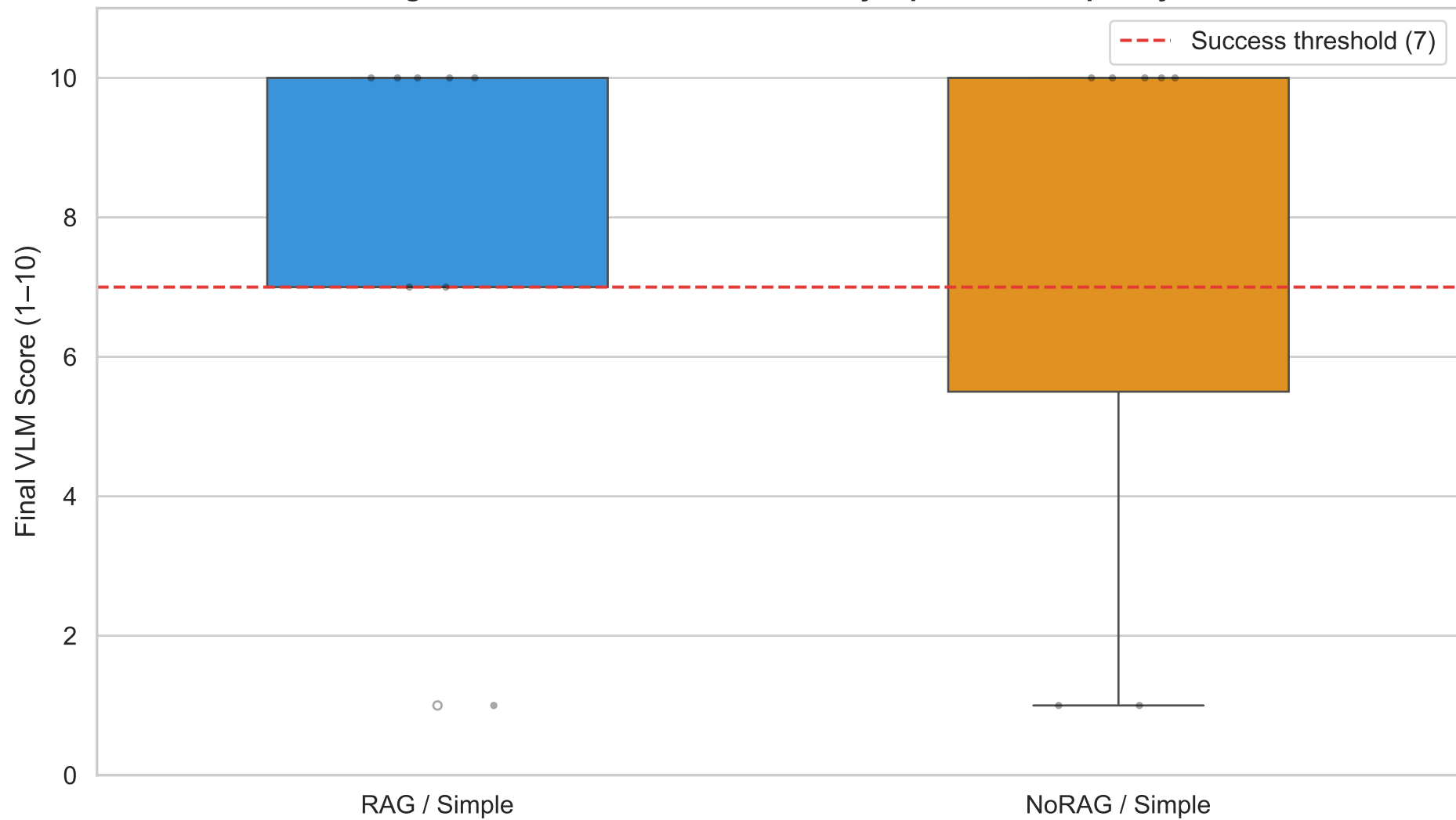


Figure 3 — Average Iterations to Reach Success

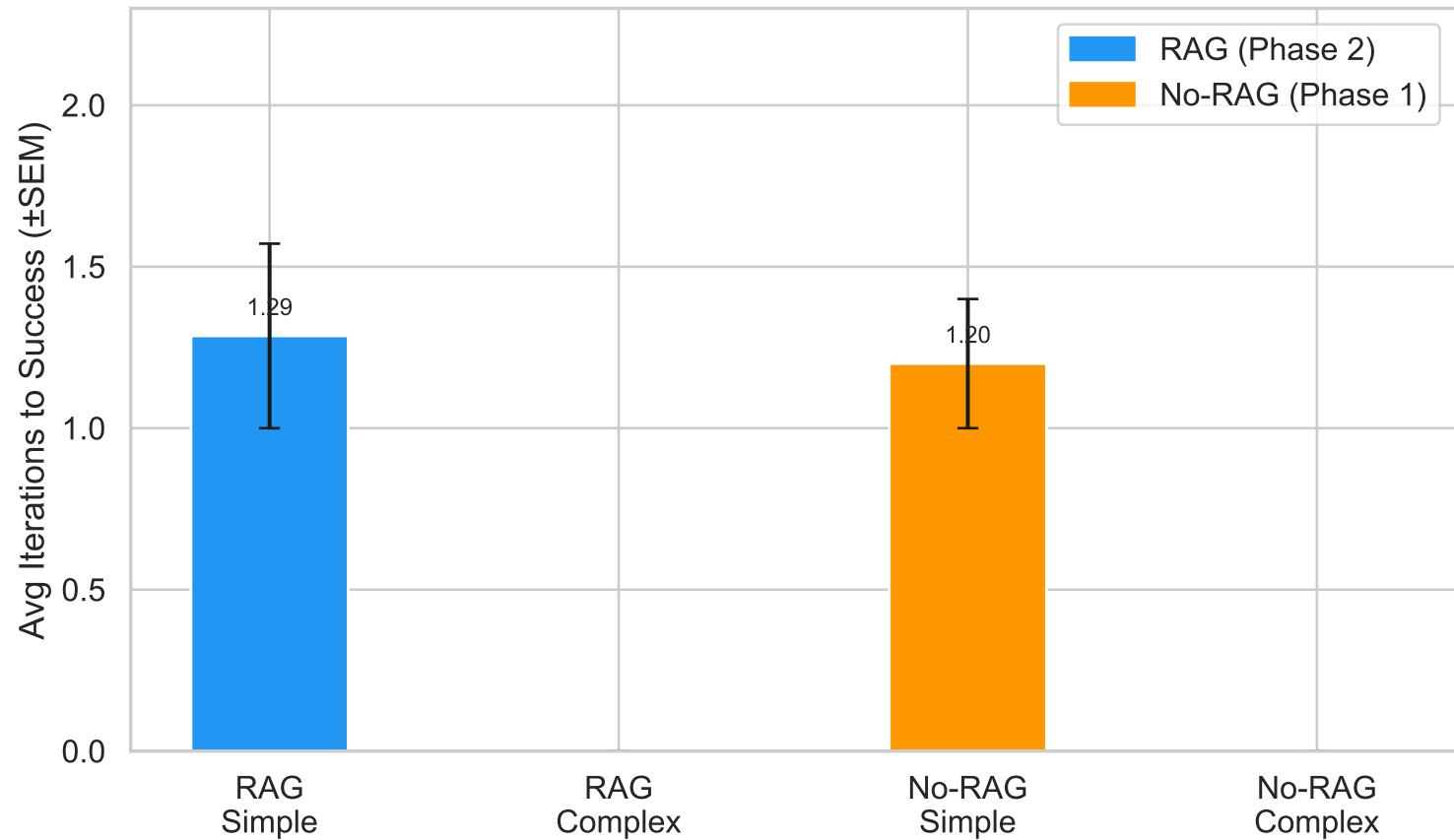


Figure 4 — Generation Time Distribution (Violin)

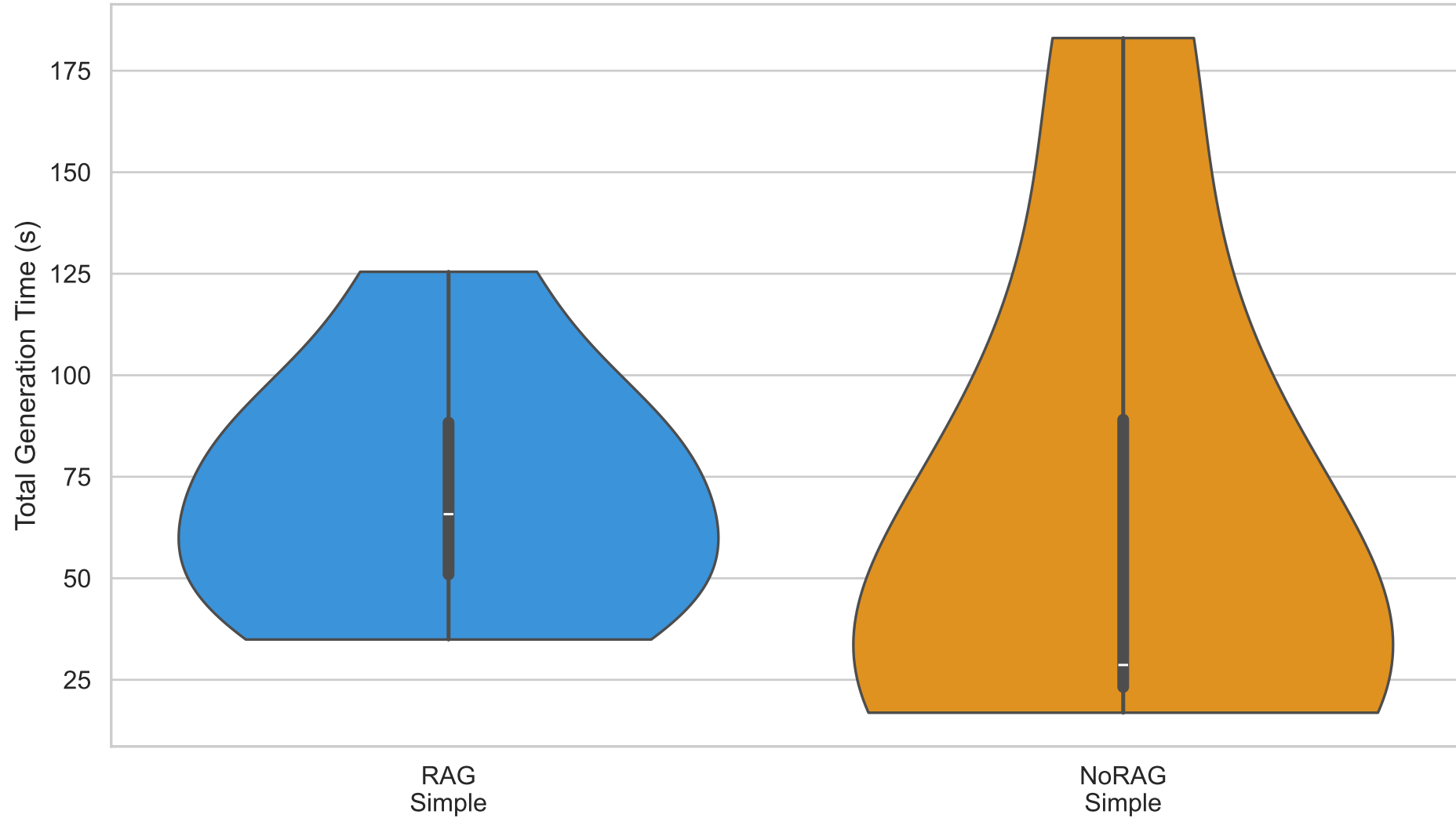


Figure 5 — First-Pass HLSL Compile Success Rate

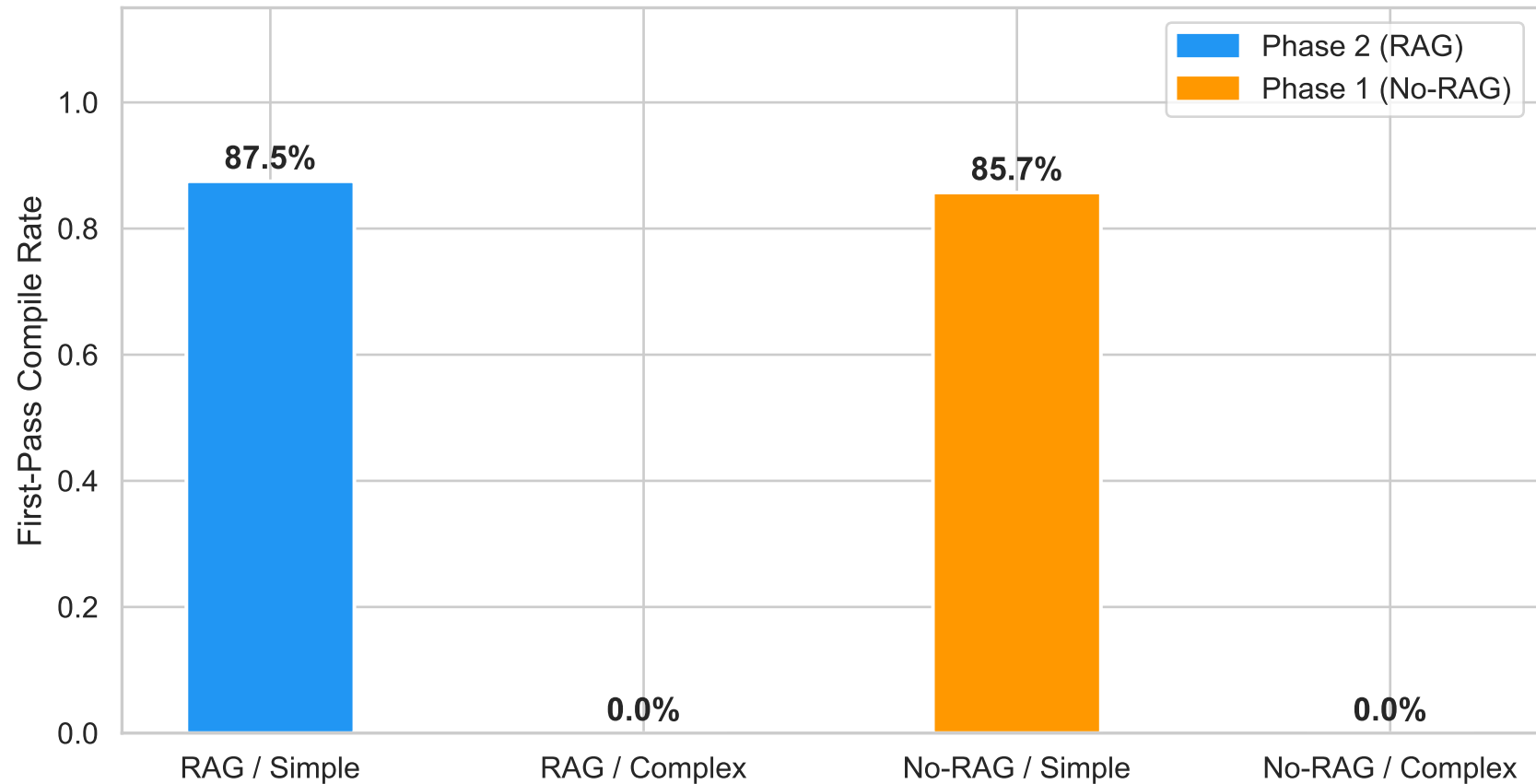


Figure 6 — VLM Score vs Generation Time

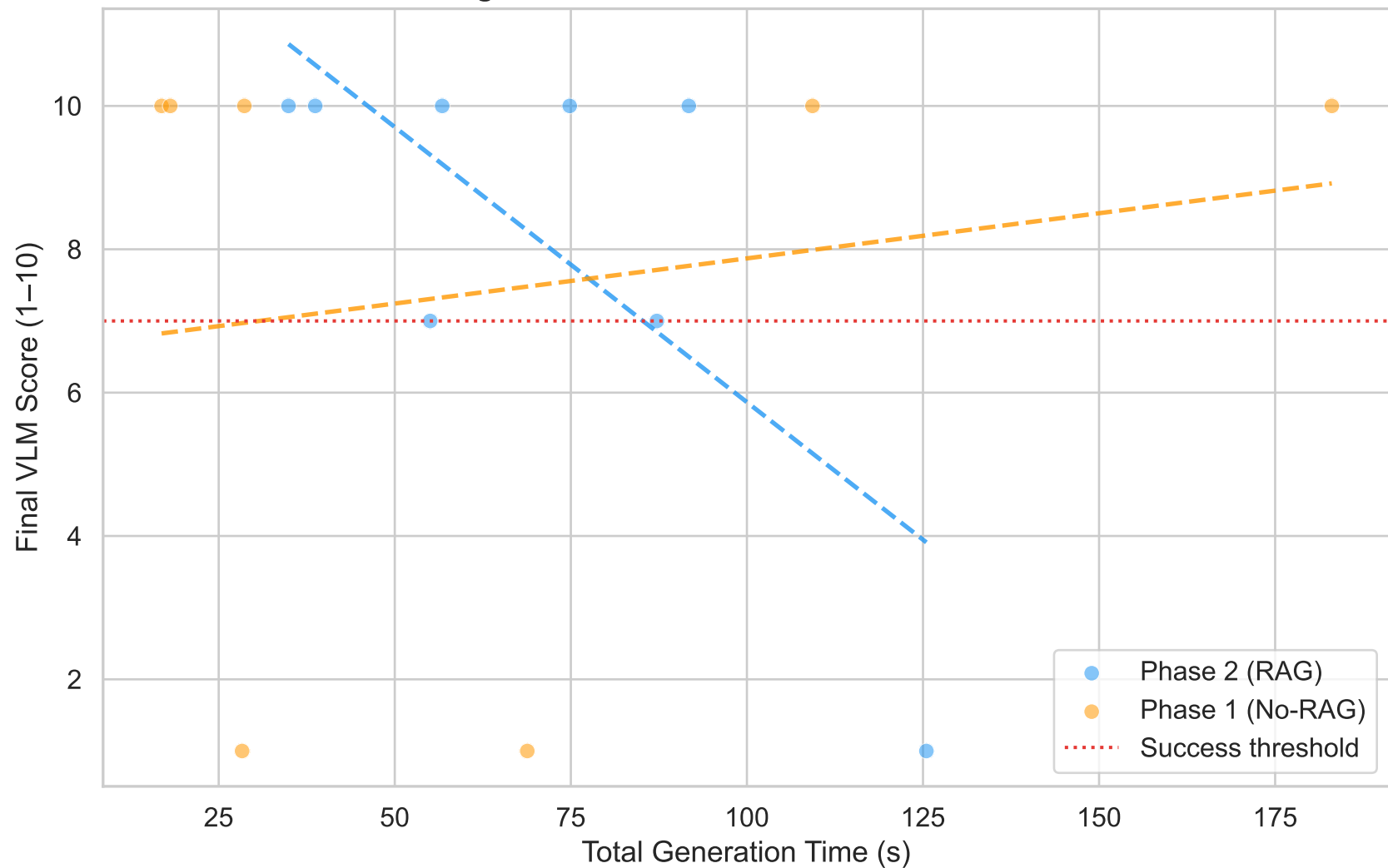


Figure 7 — VLM Score Frequency Distribution

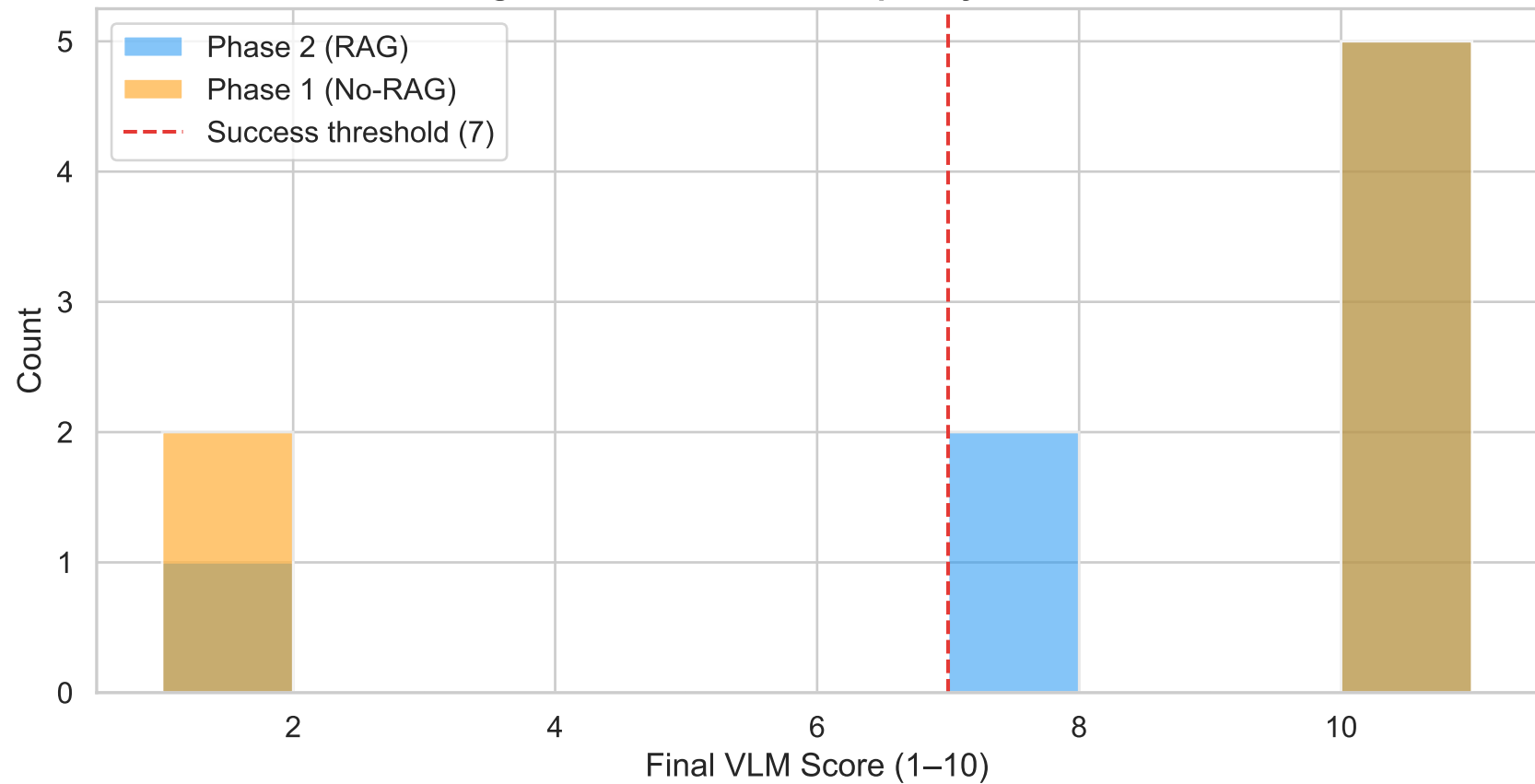


Figure 8 — Impact of Knowledge-Base Coverage (RAG Only)

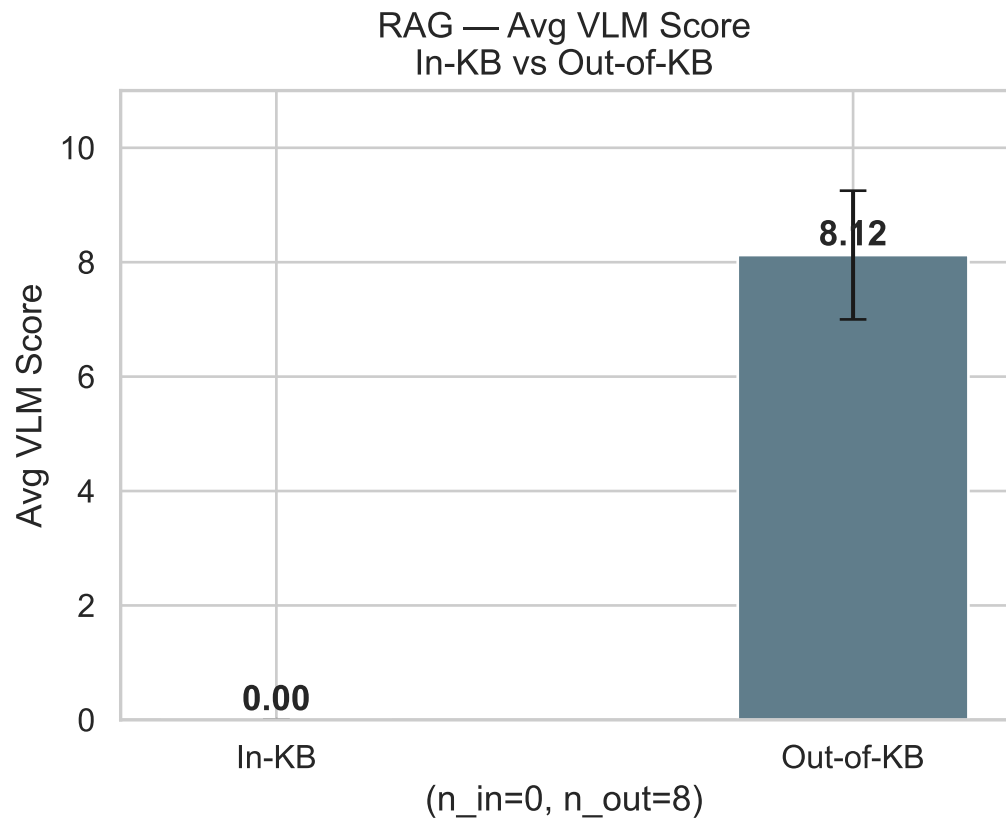
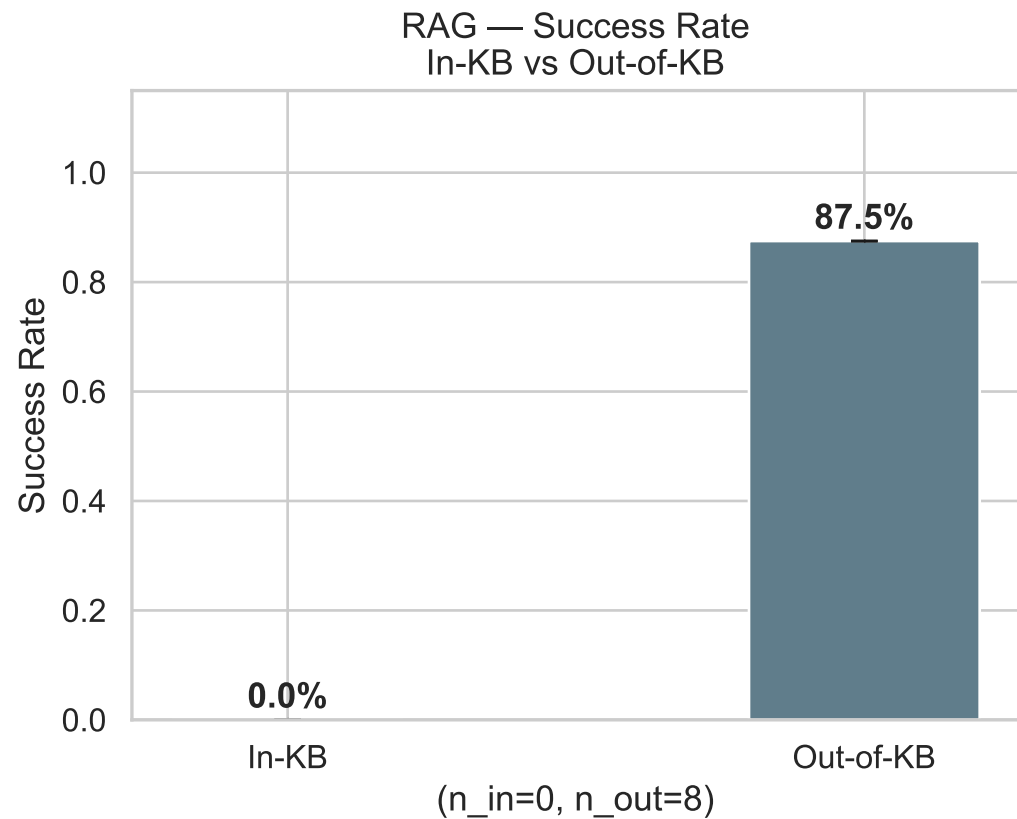


Figure 9 — Retrieval Similarity vs Generation Quality (RAG)

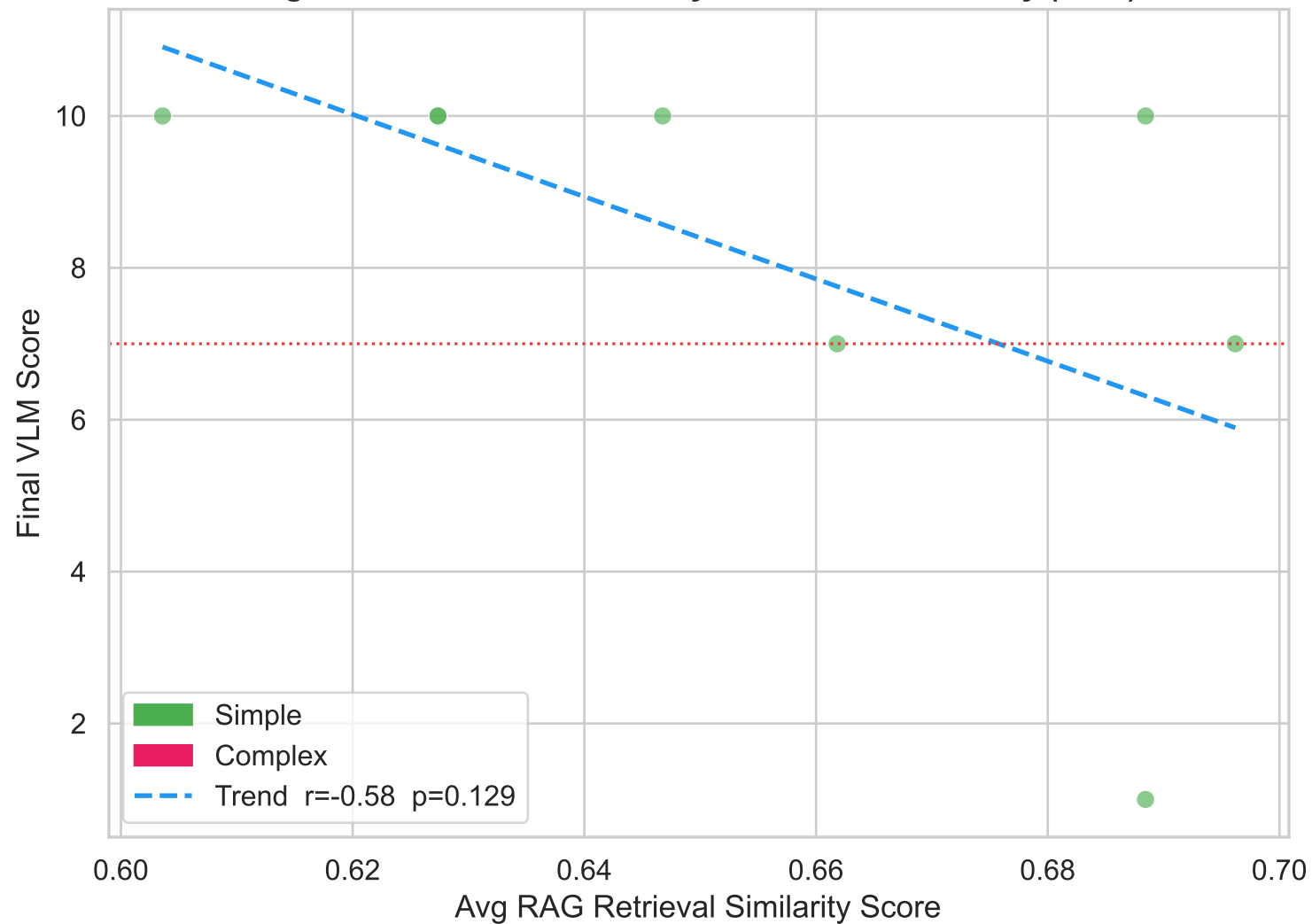


Figure 10 — RAG: Decomposition Complexity vs Success Rate

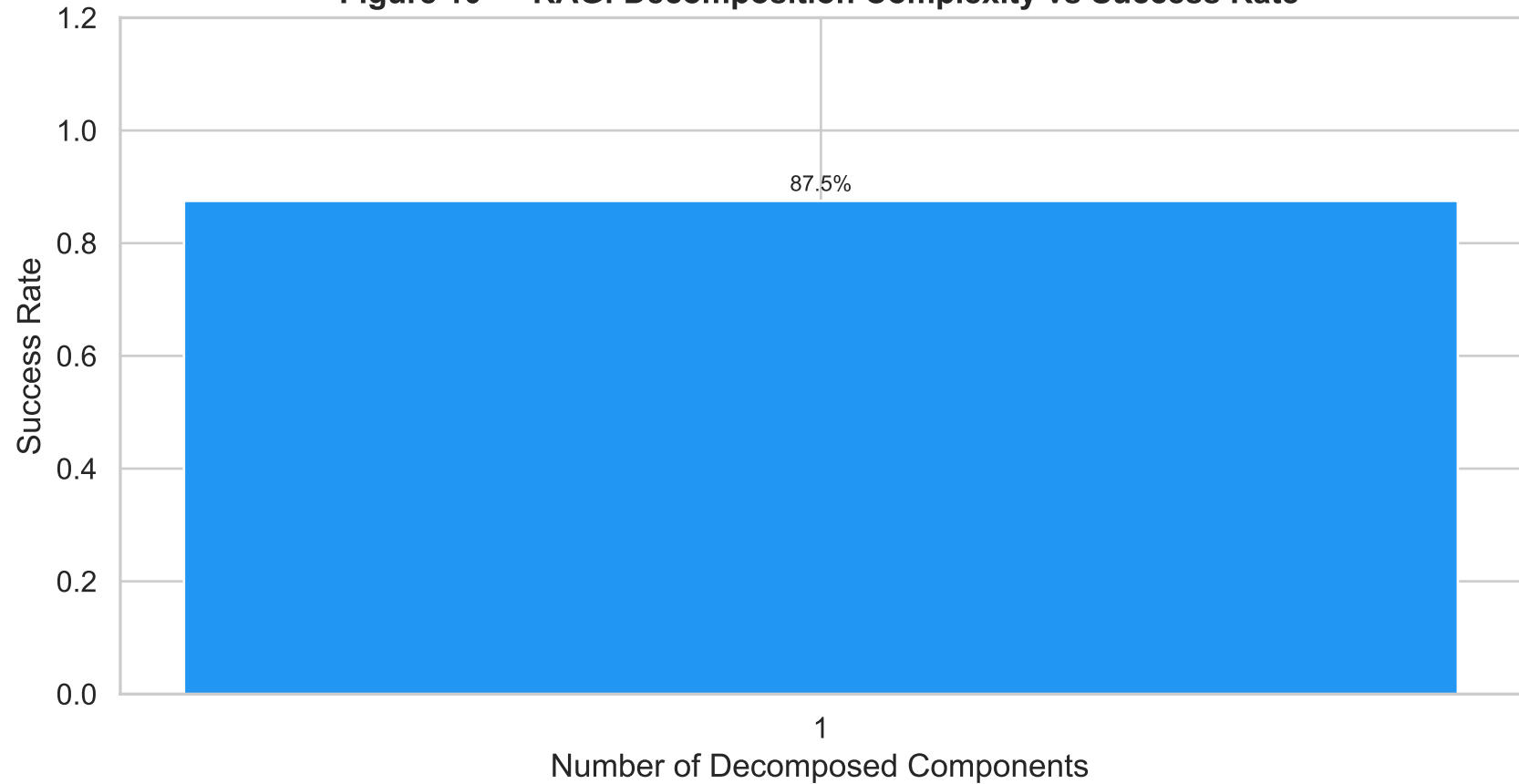


Figure 11 — Score Convergence Across Iterations

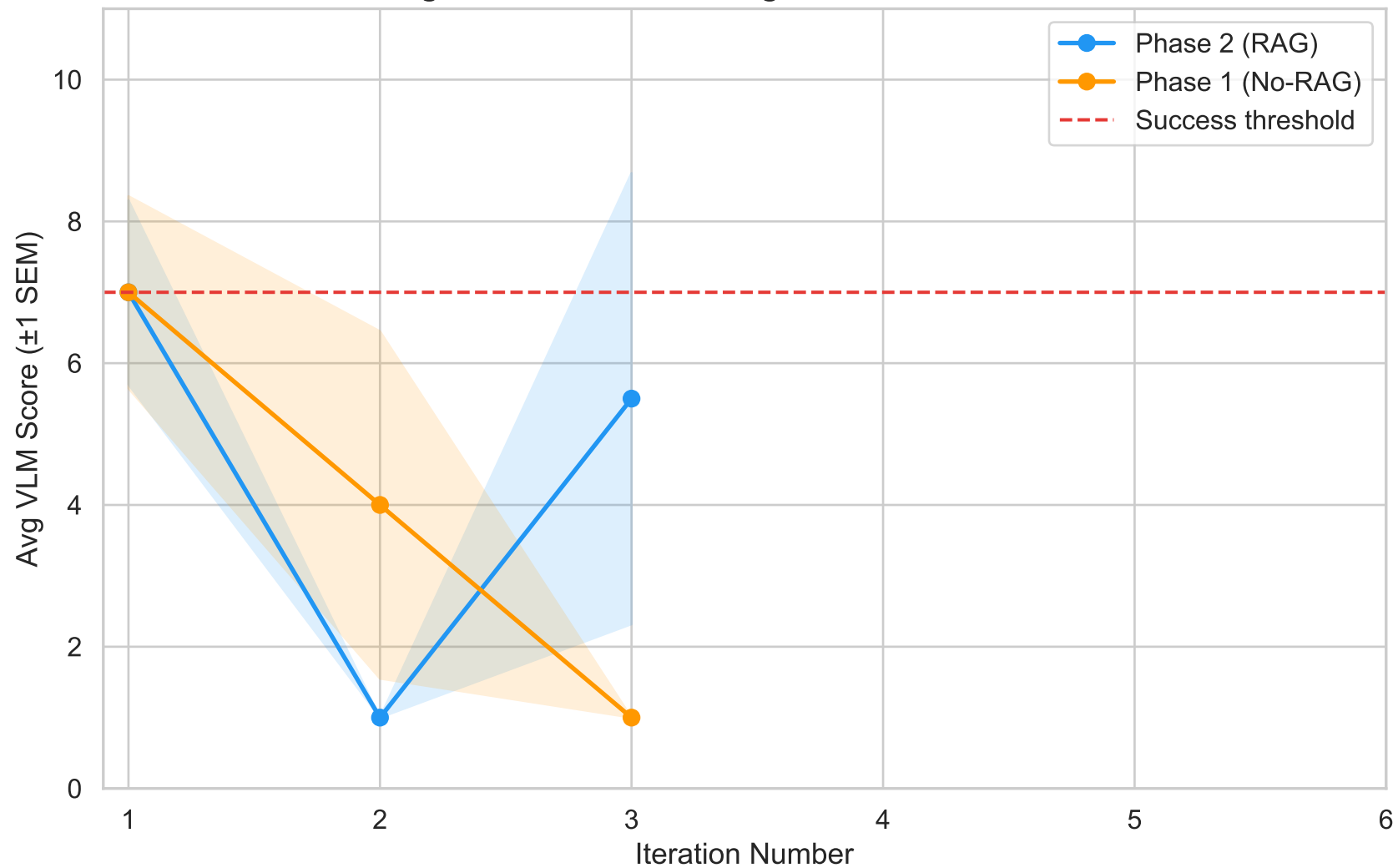


Figure 12 — Success Rate by Code LLM × Eval Provider

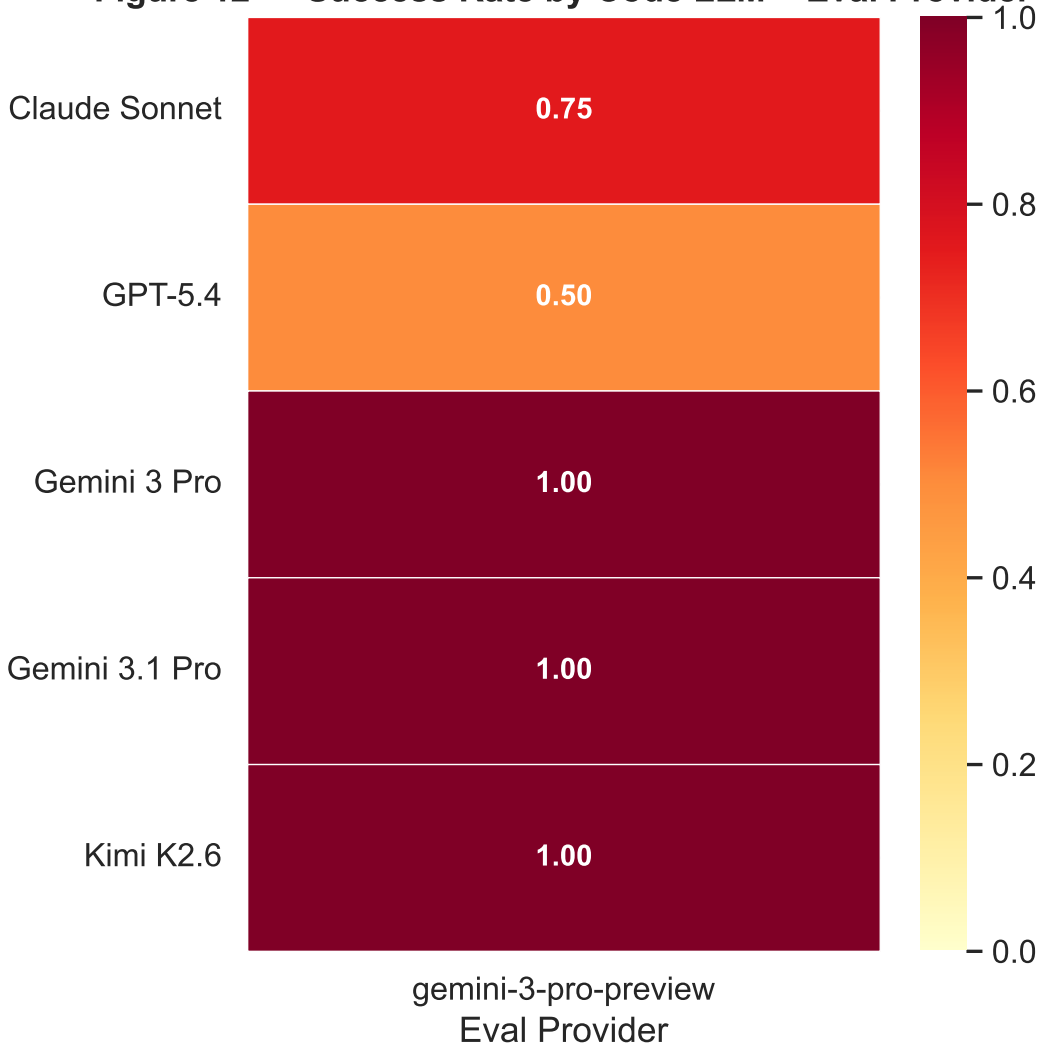


Figure 13 — VLM Score vs Human Evaluation

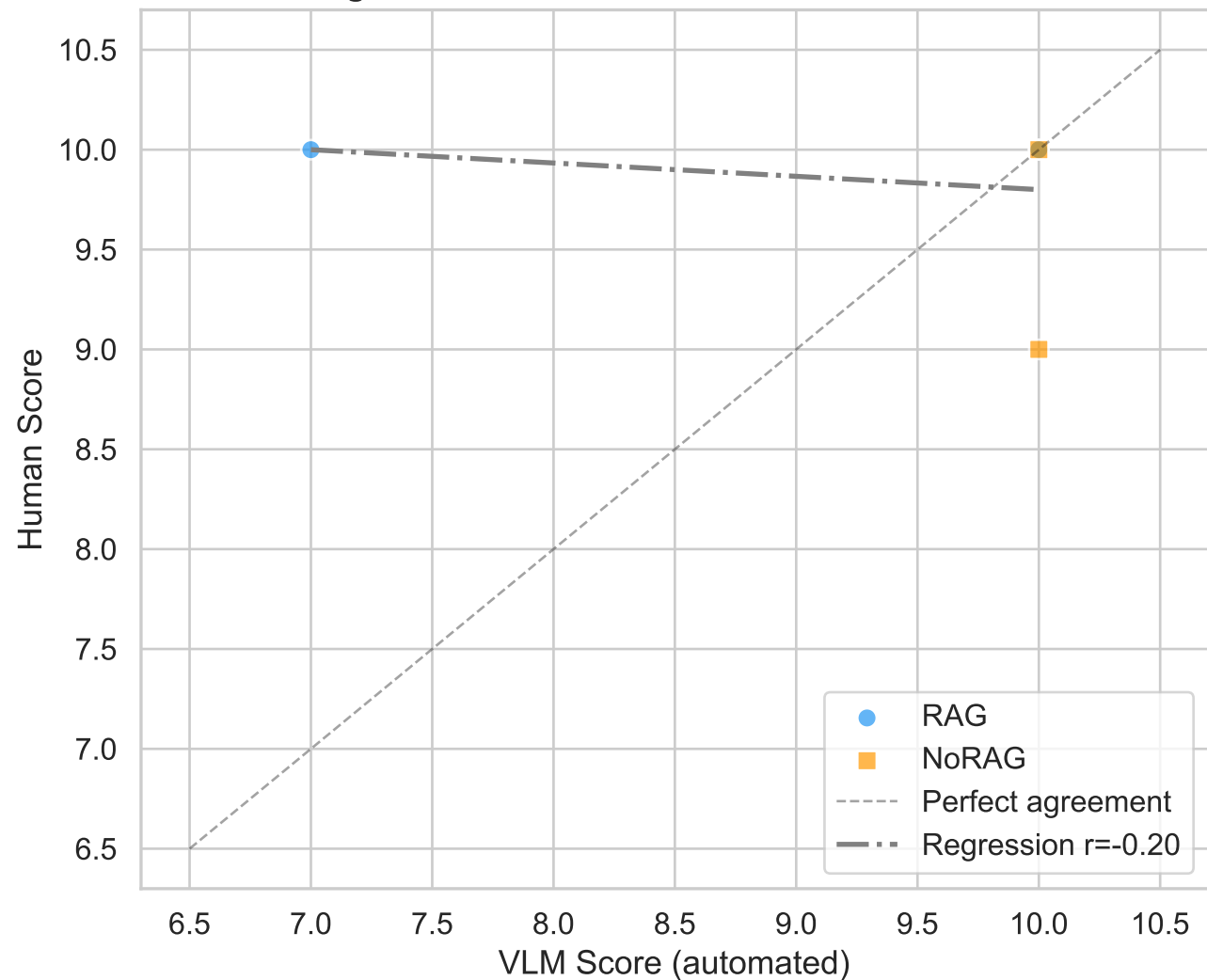


Figure 14 — Phase 1 vs Phase 2 Key Metrics Summary

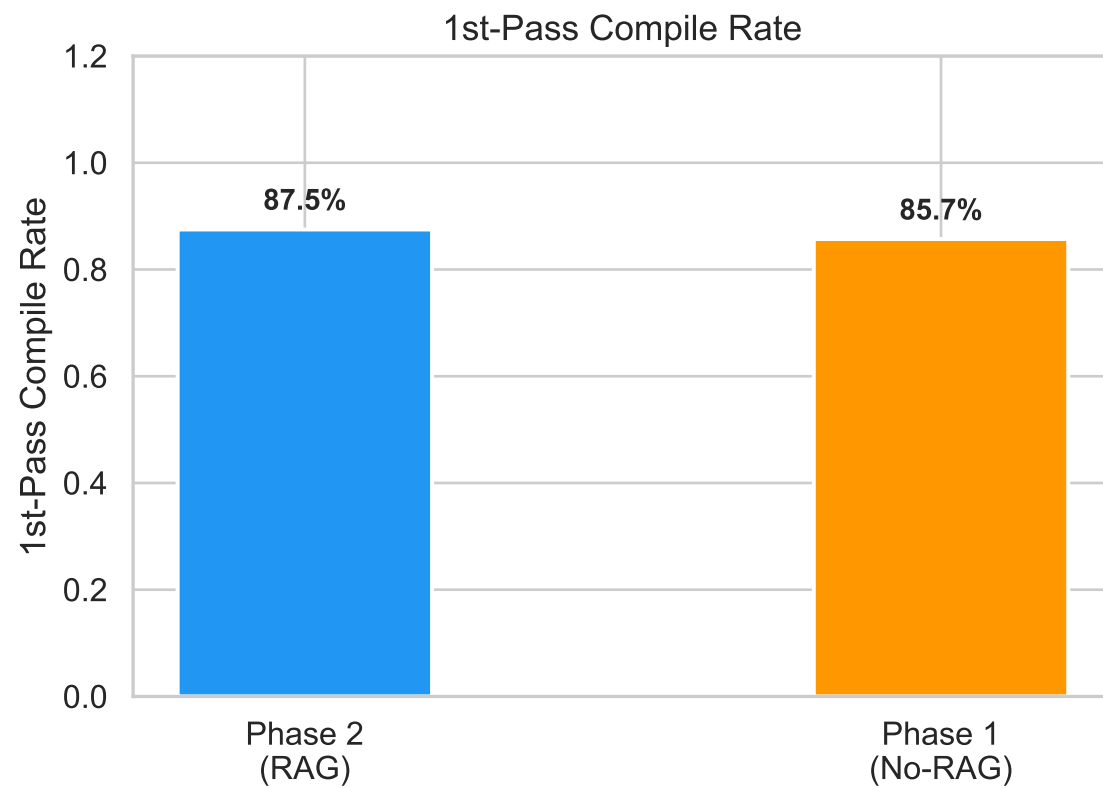
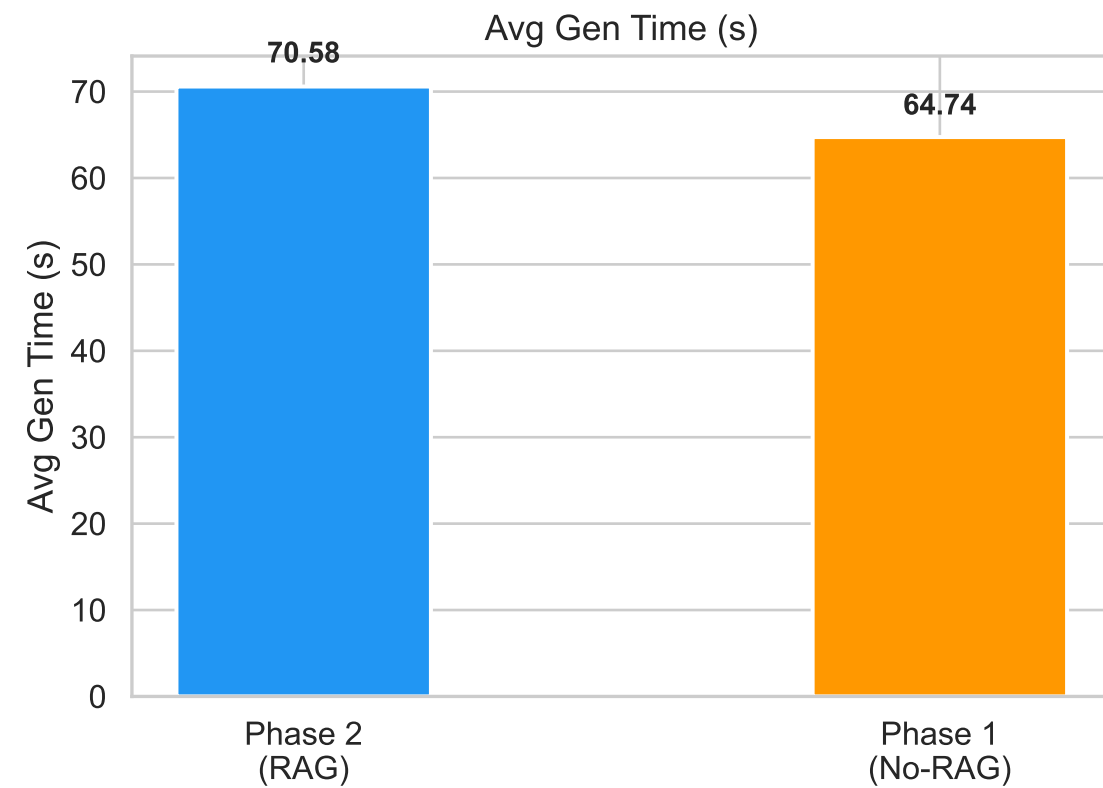
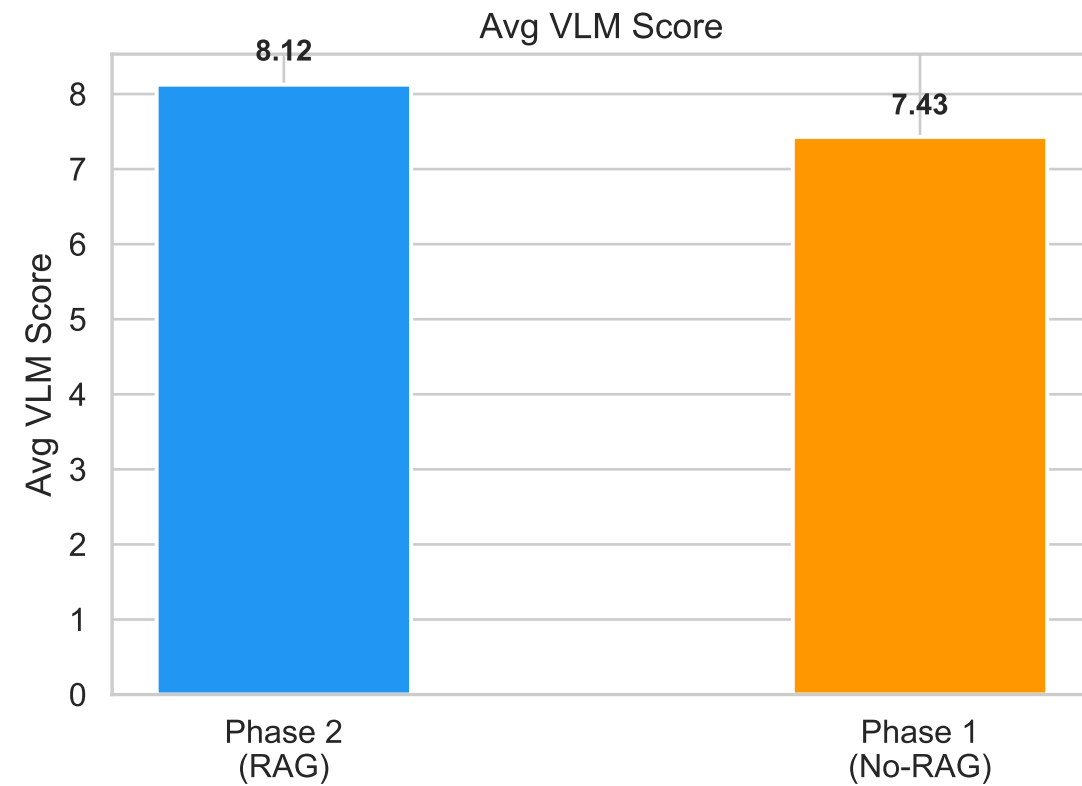
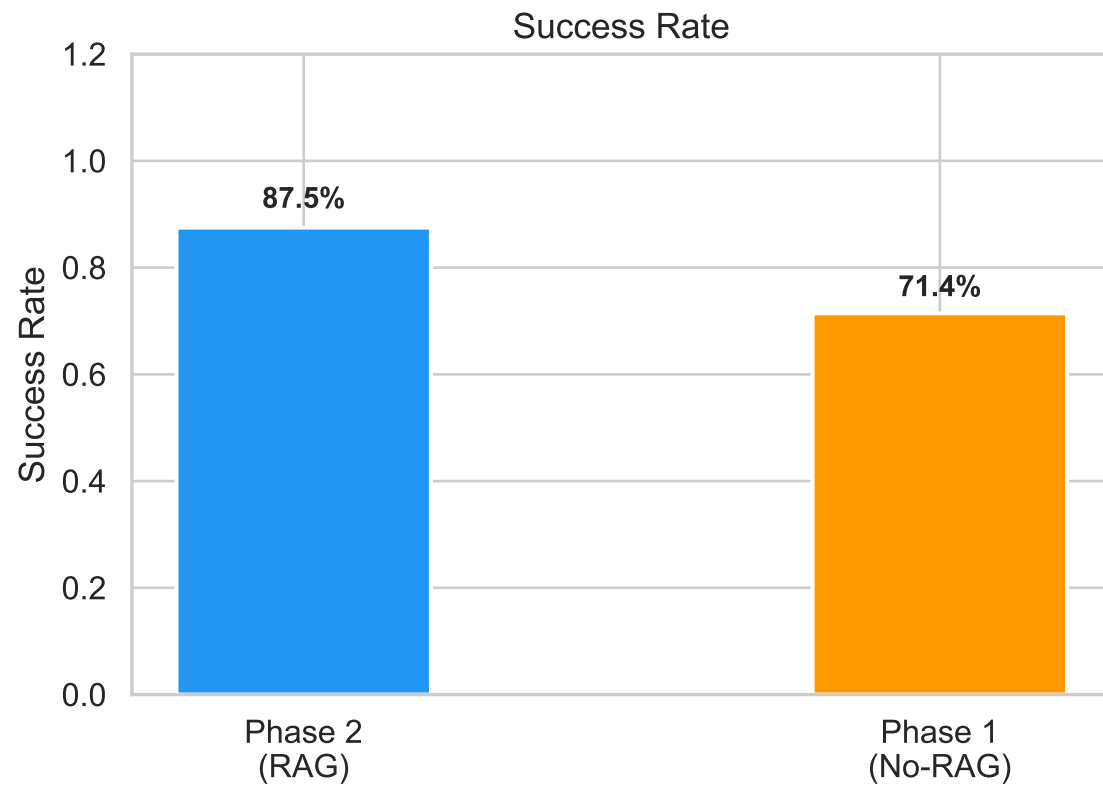
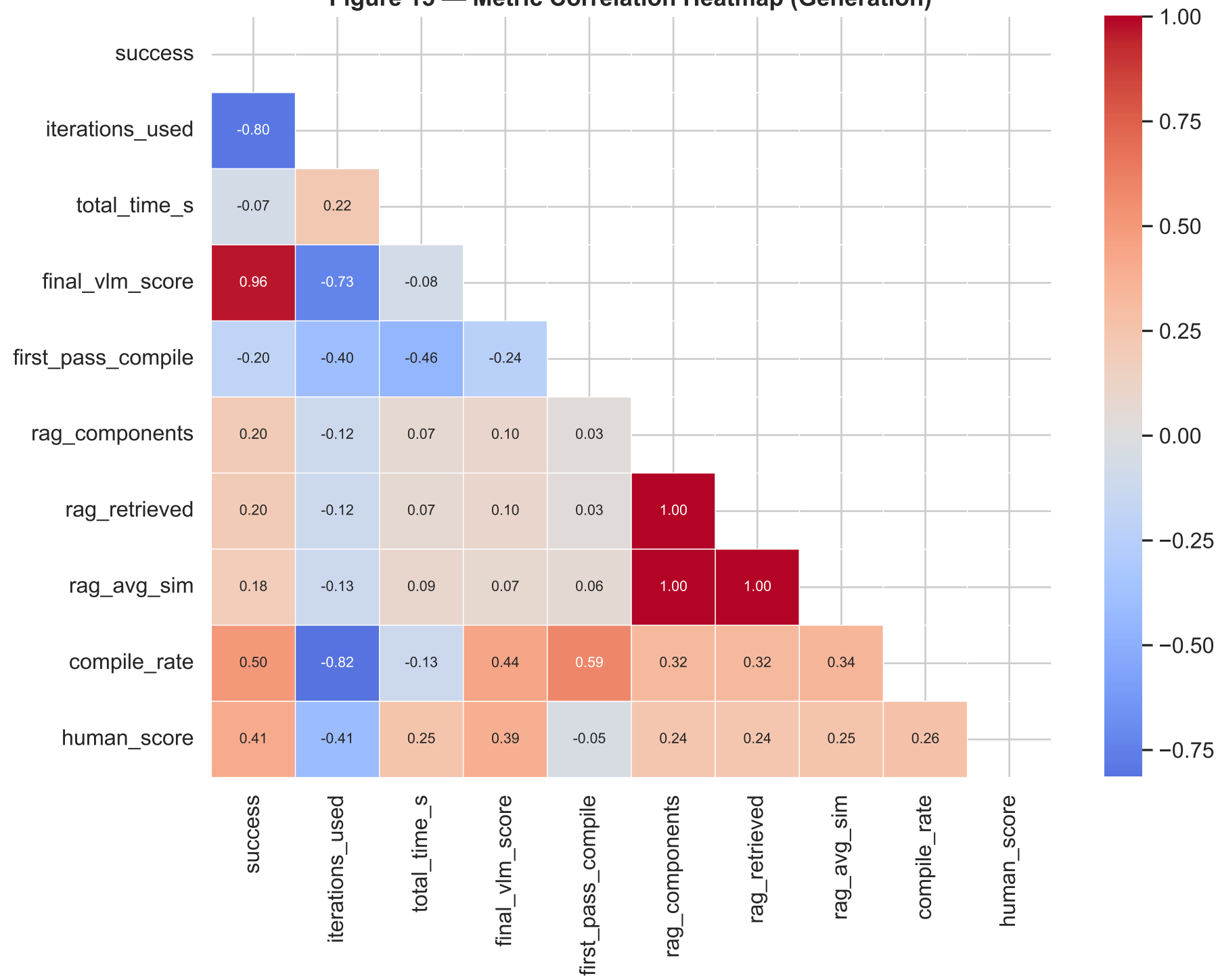


Figure 15 — Metric Correlation Heatmap (Generation)



**Figure 16 — Radar: Phase 1 vs Phase 2
Normalised Metrics**

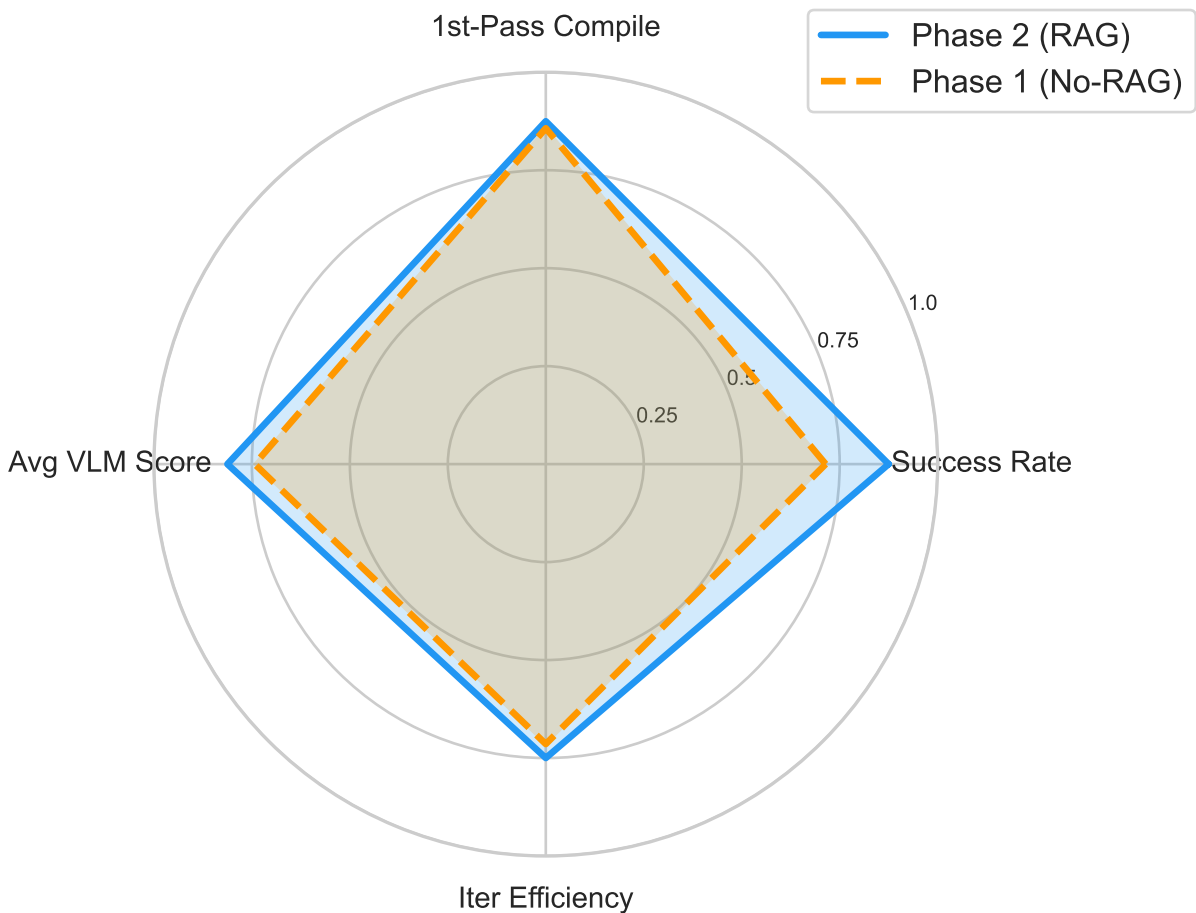


Figure 23 — Overall Success Rate: All Experiment Types

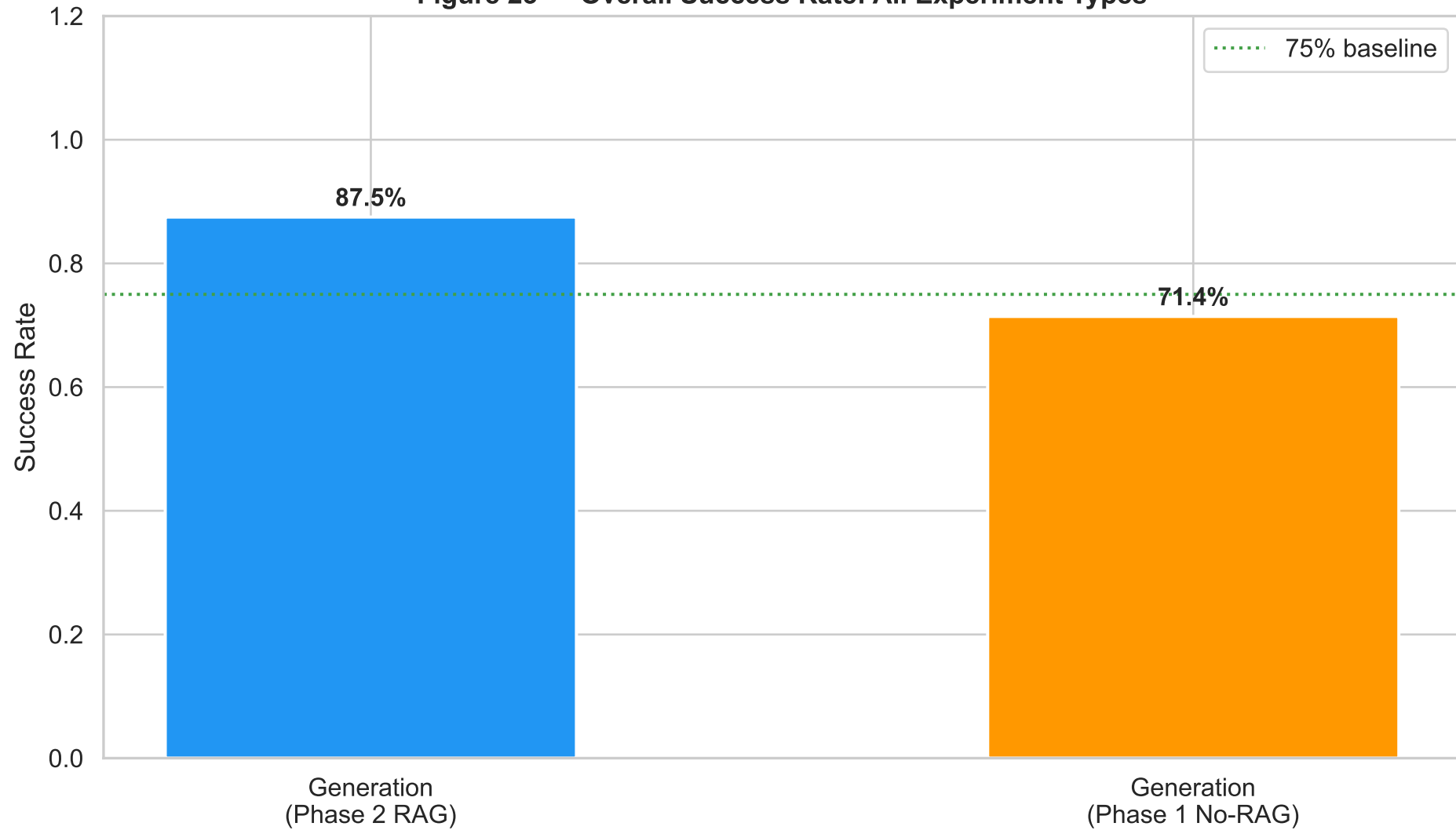


Figure 24 — Success Rate by LLM Model & Pipeline

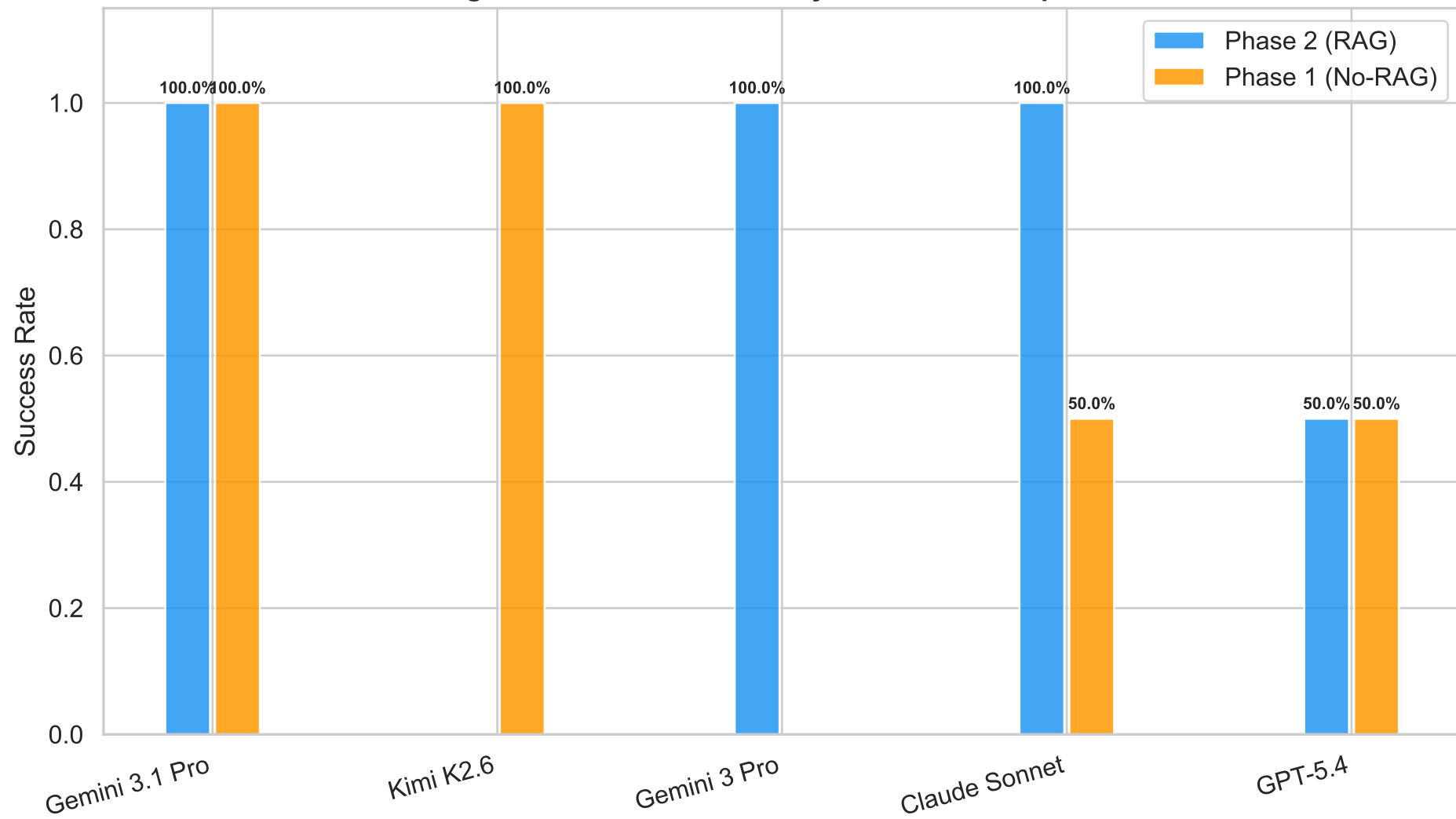


Figure 25 — VLM Score Distribution by LLM Model

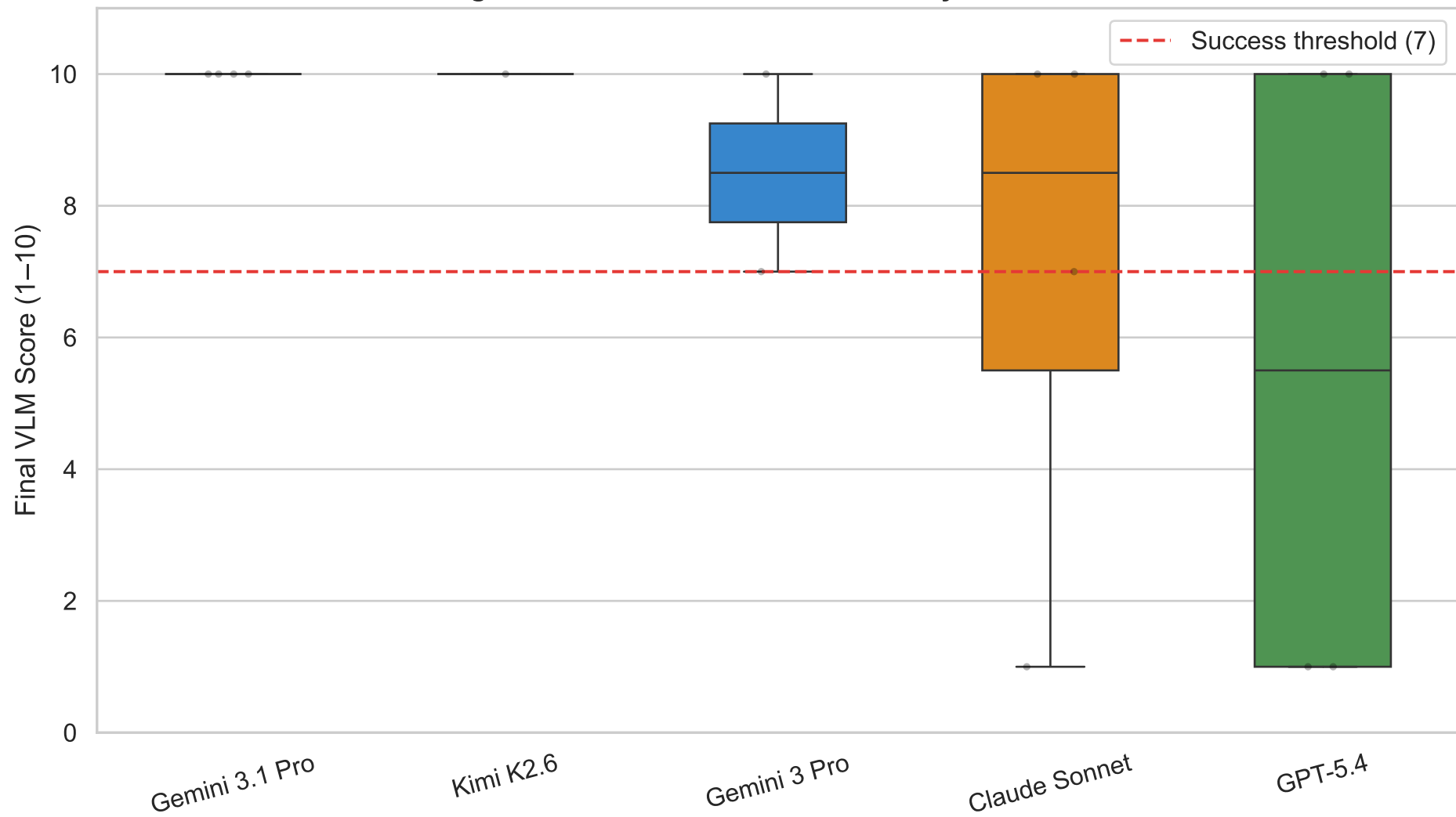


Figure 26 — First-Pass HLSL Compile Rate by LLM Model

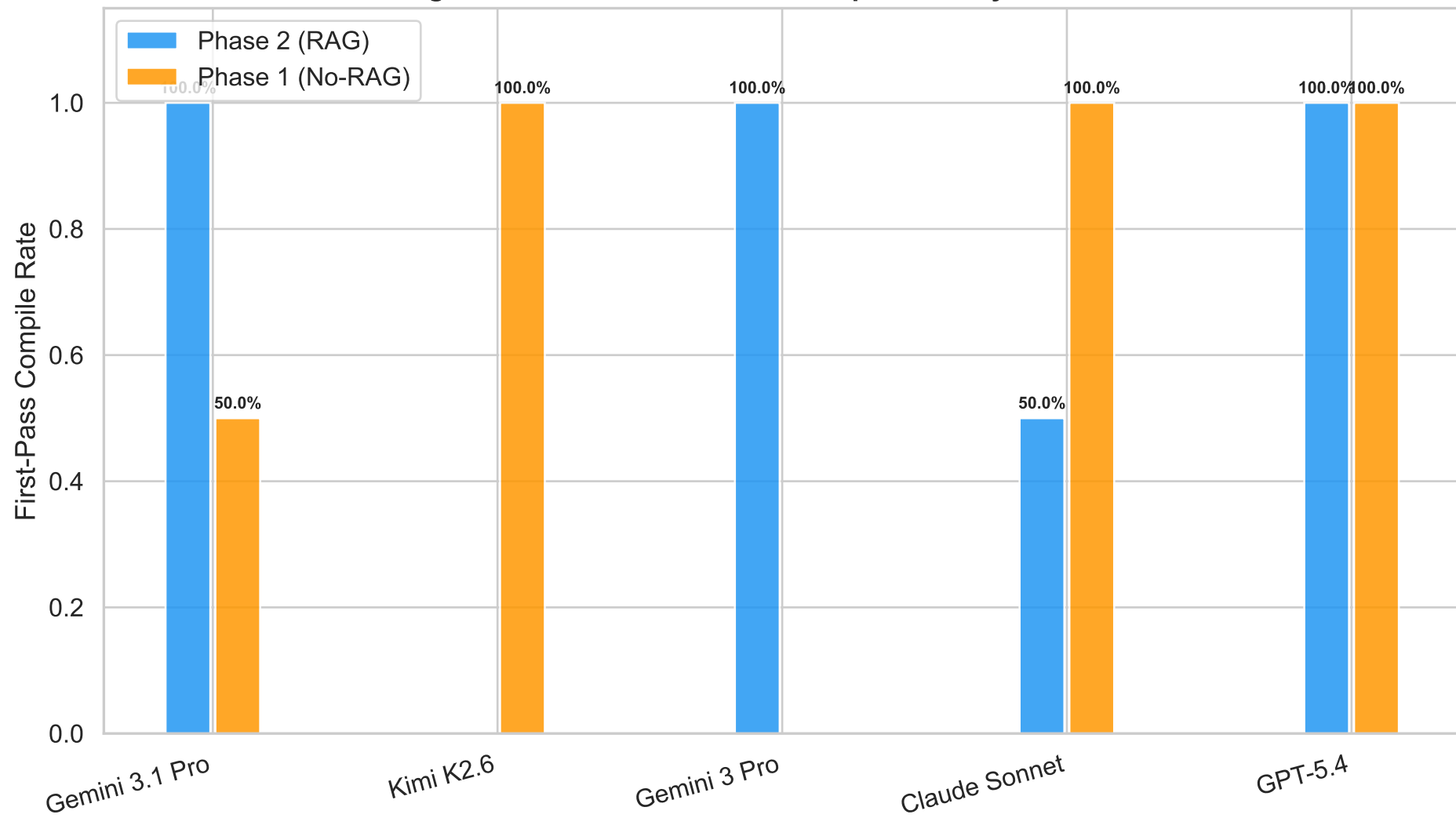


Figure 27 — Average Iterations to Success by LLM Model

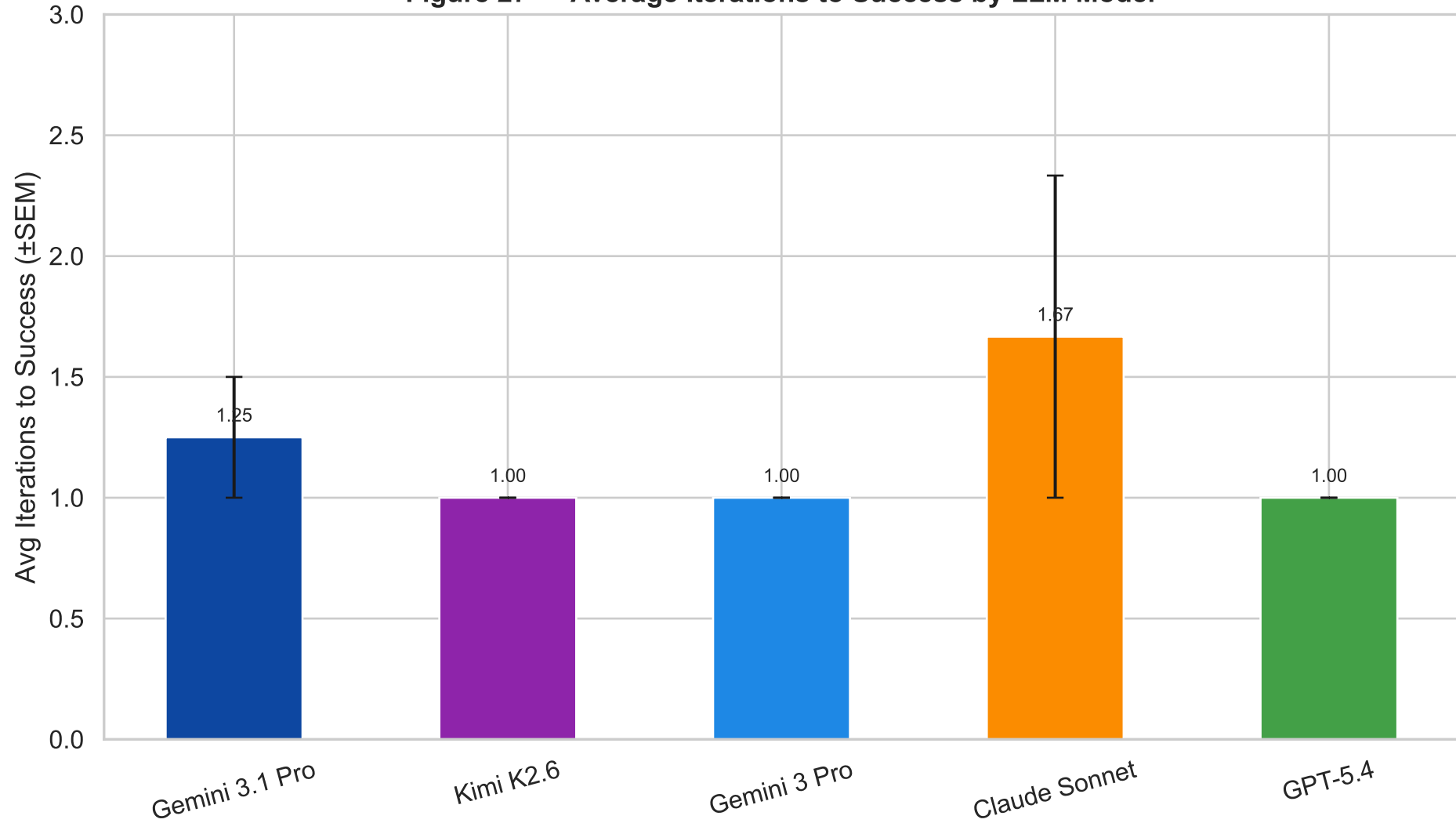


Figure 28 — Generation Time by LLM Model

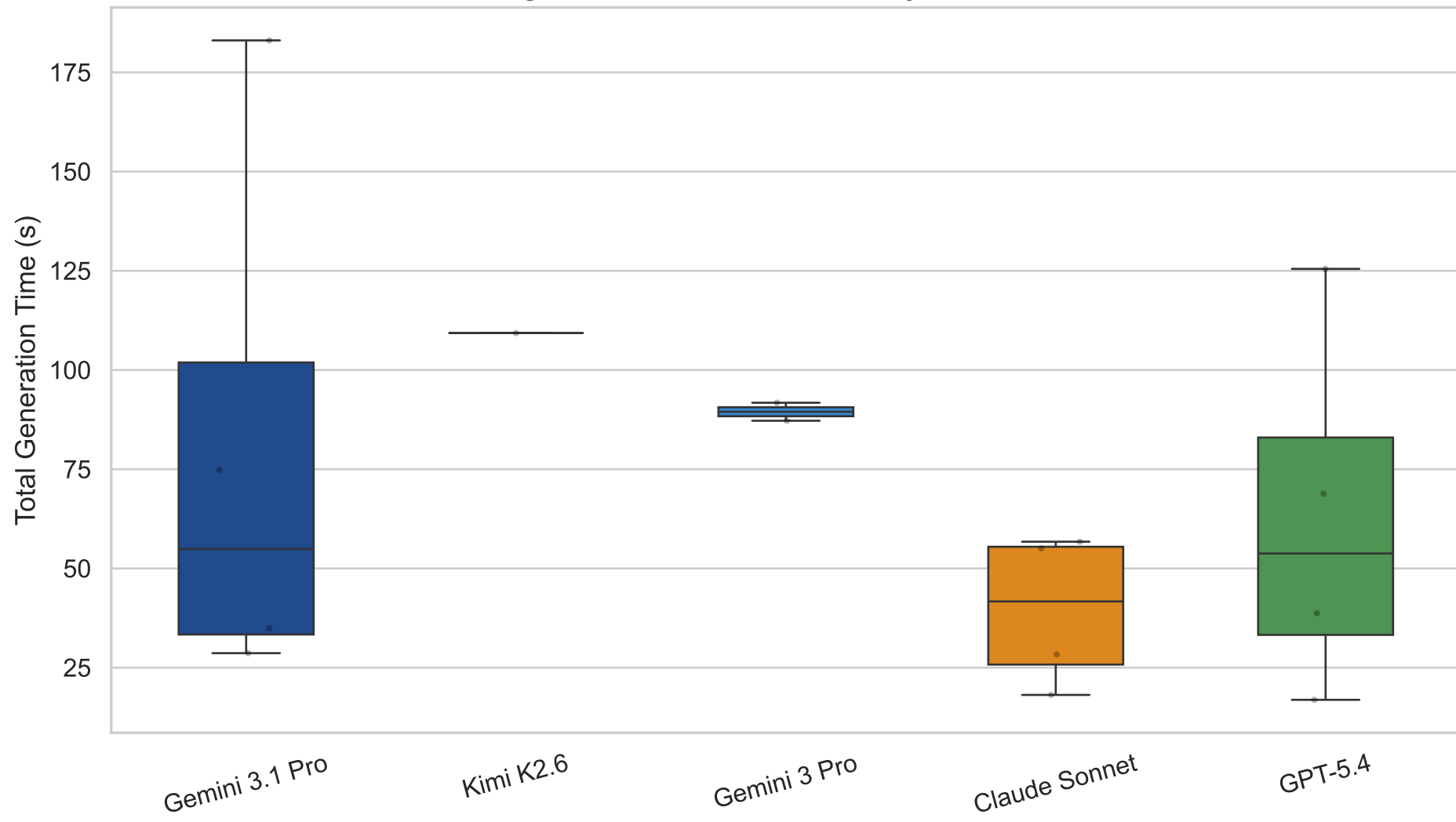


Figure 29 — Cost & Token Usage by LLM Model

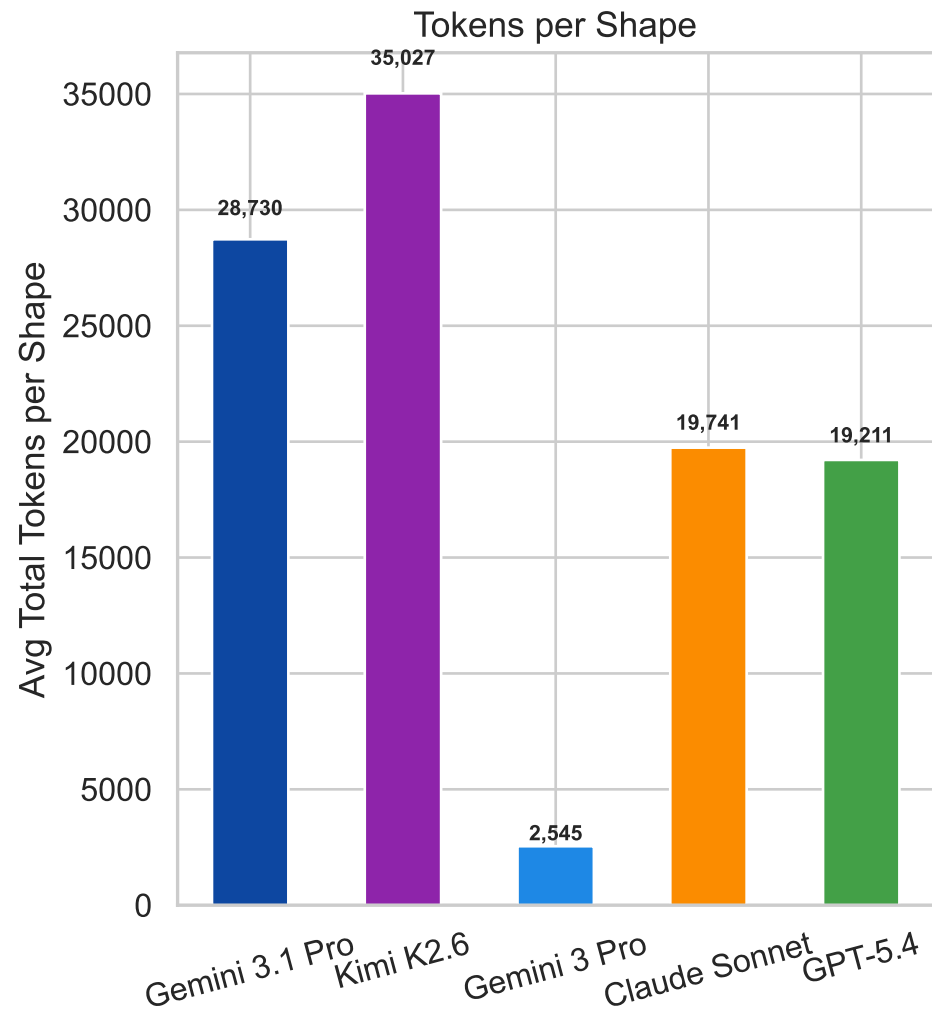
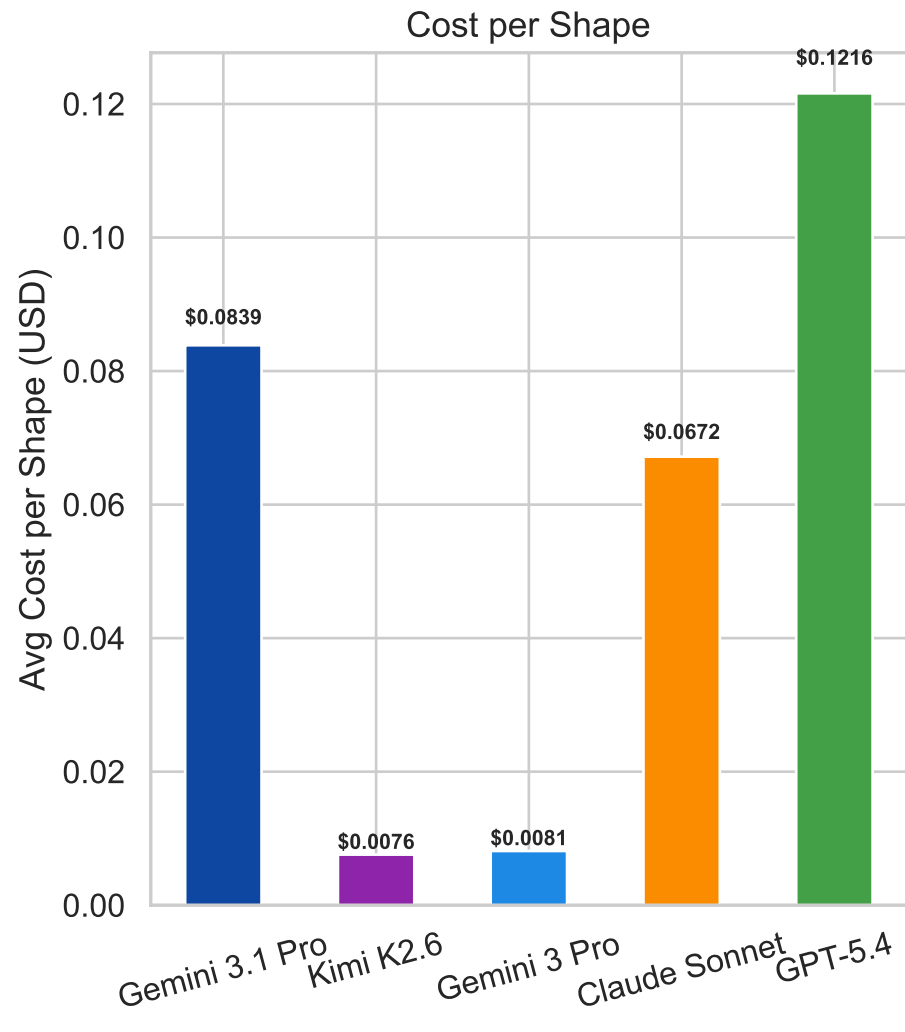


Figure 30 — HLSL Compile Rate per Iteration by LLM Model

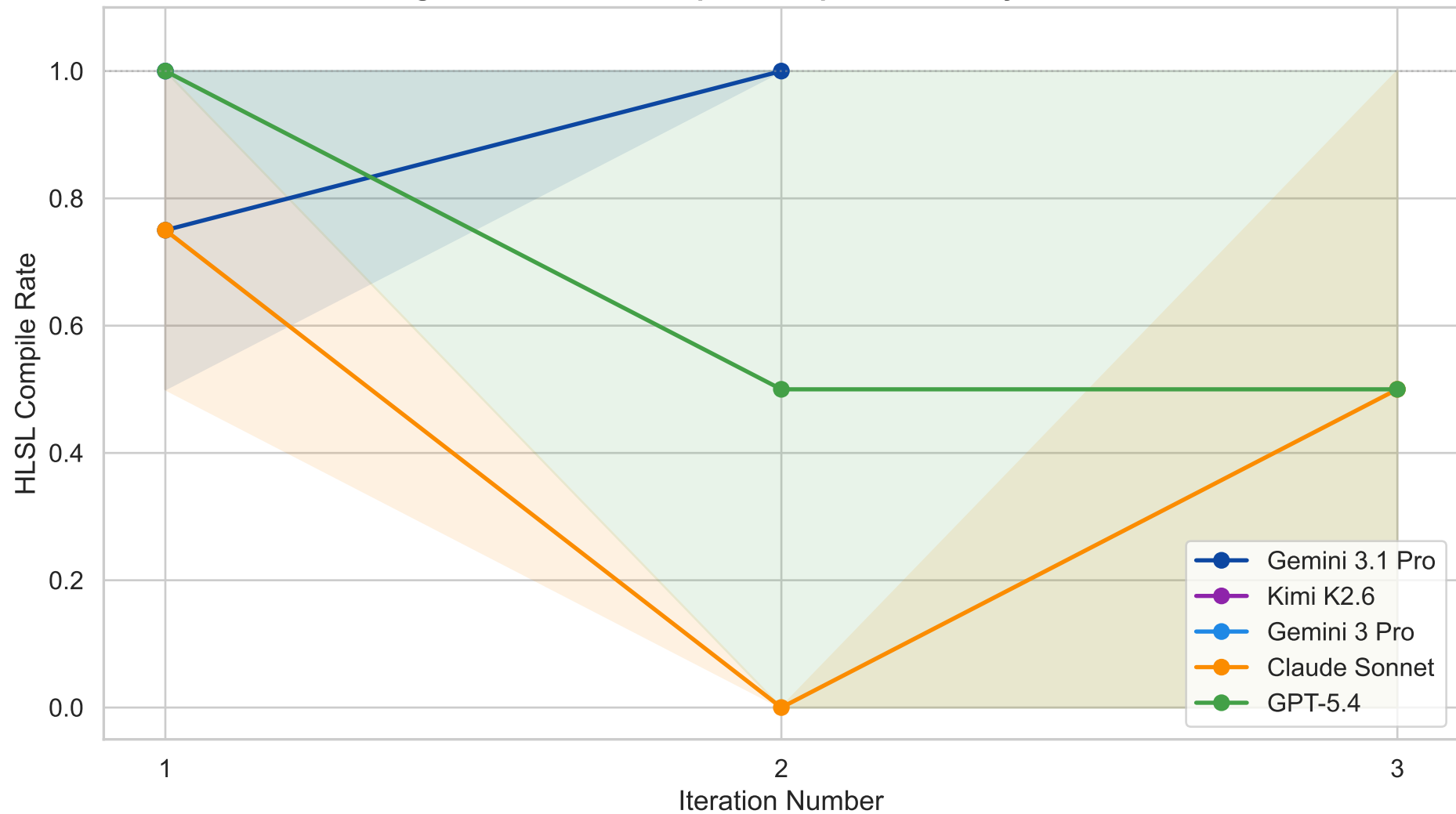


Figure 31 — VLM Score Convergence per Iteration by LLM Model

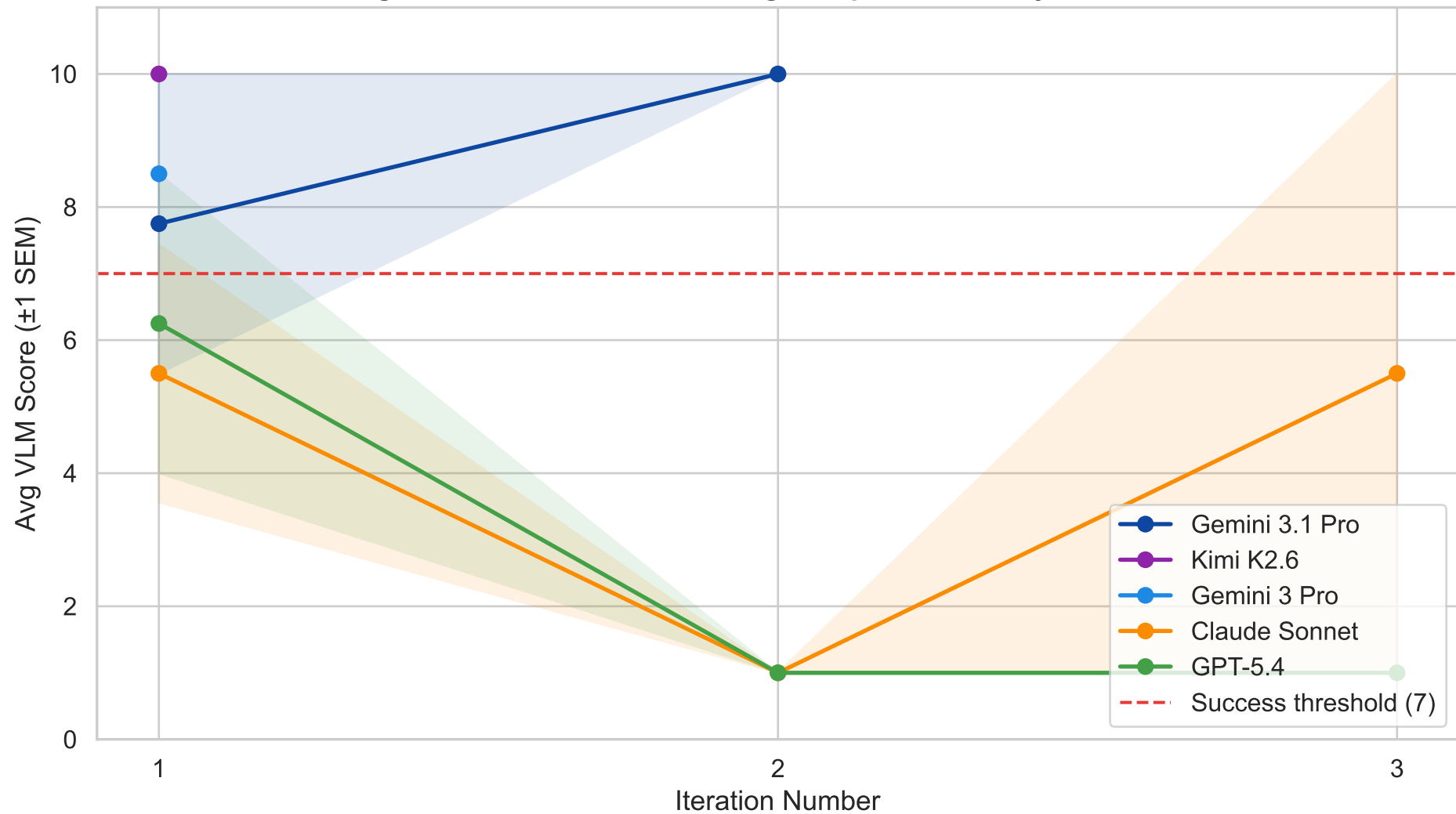


Figure 32 — Iteration Recovery Rate by LLM Model
(Shapes where Iter 1 failed to compile — did Iter 2 compile?)

